

Fasting-only Principal Component Analysis

Data scope

- Conditions retained: Baseline (2); NS4 Fasting (18); RTDS Fasting (26).
- NS4 after-meal rows excluded before PCA: 32.
- Fasting rows available: 95; complete PCA rows retained: 67.
- PCA rows excluded (missing/non-positive): 28.

PCA specification

- Variables: 15 measured bile acids only; aggregate classes were not included.
- Preprocessing: LTR median plate adjustment, natural-log transform, centering, and unit-variance scaling.
- Missing values: complete-case matrix across the 15 bile acids; no imputation.
- Algorithm: R prcomp using singular value decomposition.

Interpretation note

PCA is an unsupervised descriptive view of fasting-only multivariate structure.

It complements the fasting mixed-model tests and does not replace those repeated-measures inferences.

PC1 variance: 40.9%

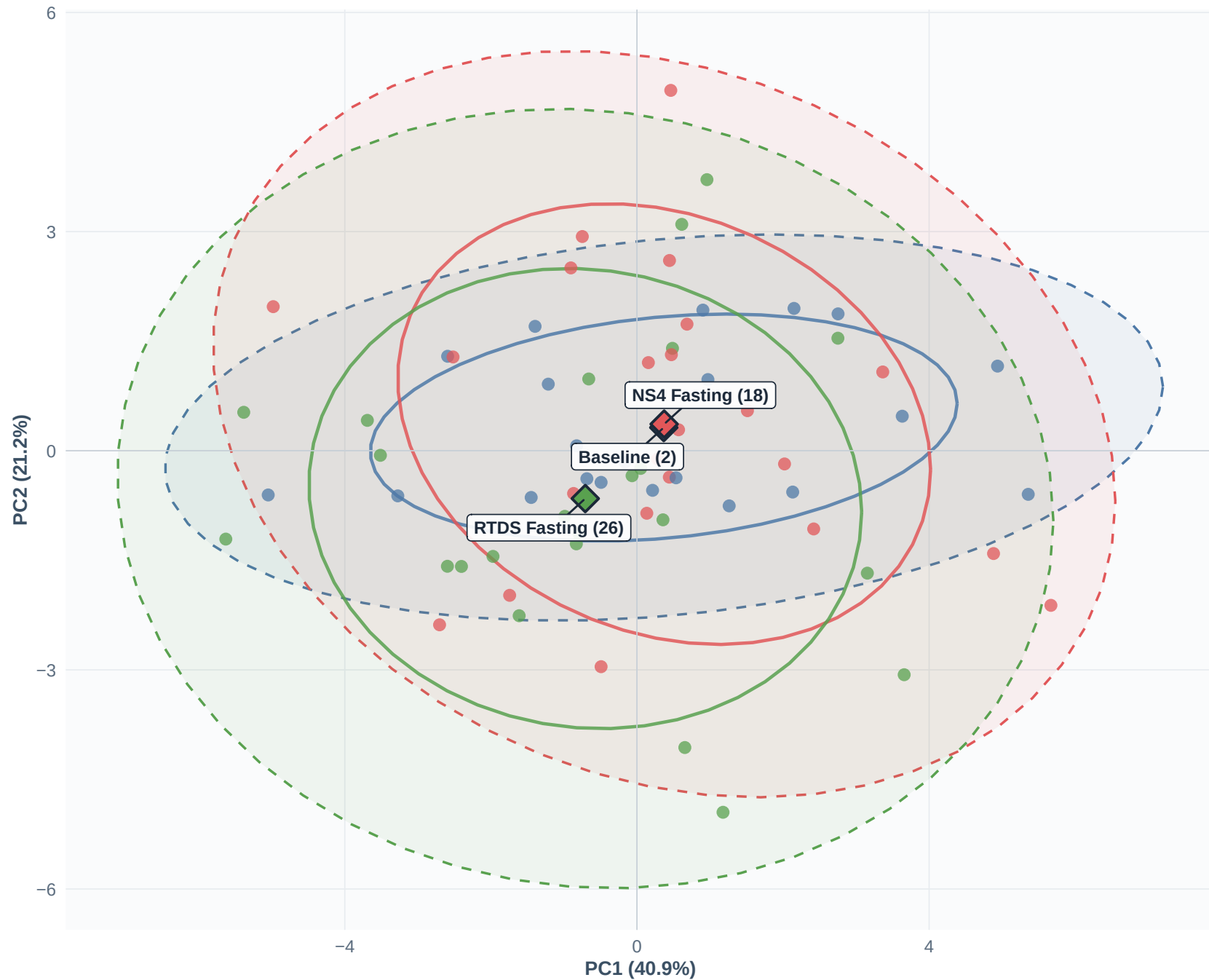
PC2 variance: 21.2%

PC1+PC2 cumulative: 62.0%

PC1+PC2+PC3 cumulative: 74.3%

PCA Score Plot — Fasting Conditions

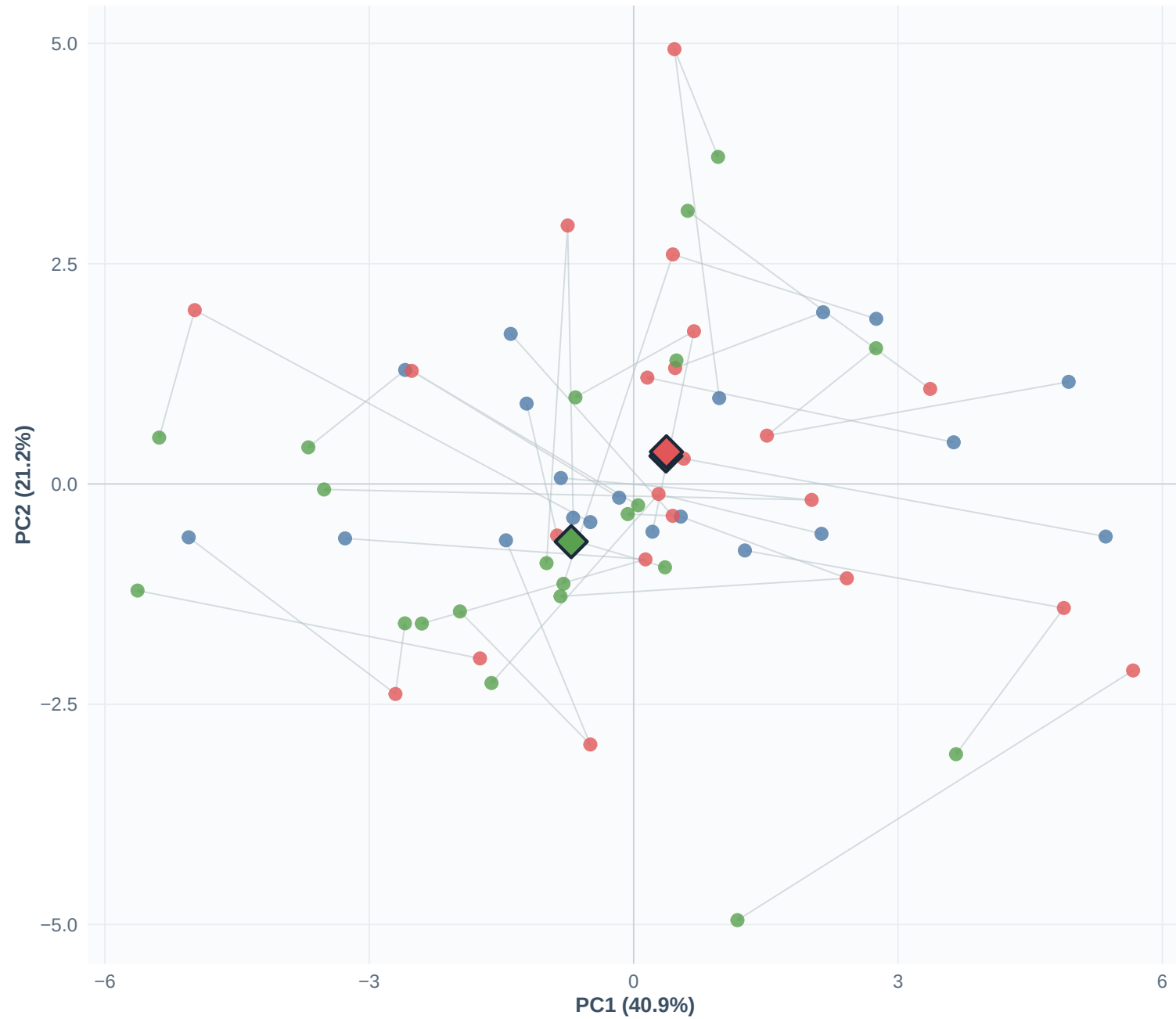
n = 67 complete observations | 27 subjects



Diamonds = group centroids | Dashed = 95% CI | Solid = 68% CI

Subject Trajectories in PCA Space

Arrows connect repeated measures per subject (n = 23 subjects with =2 timepoints)

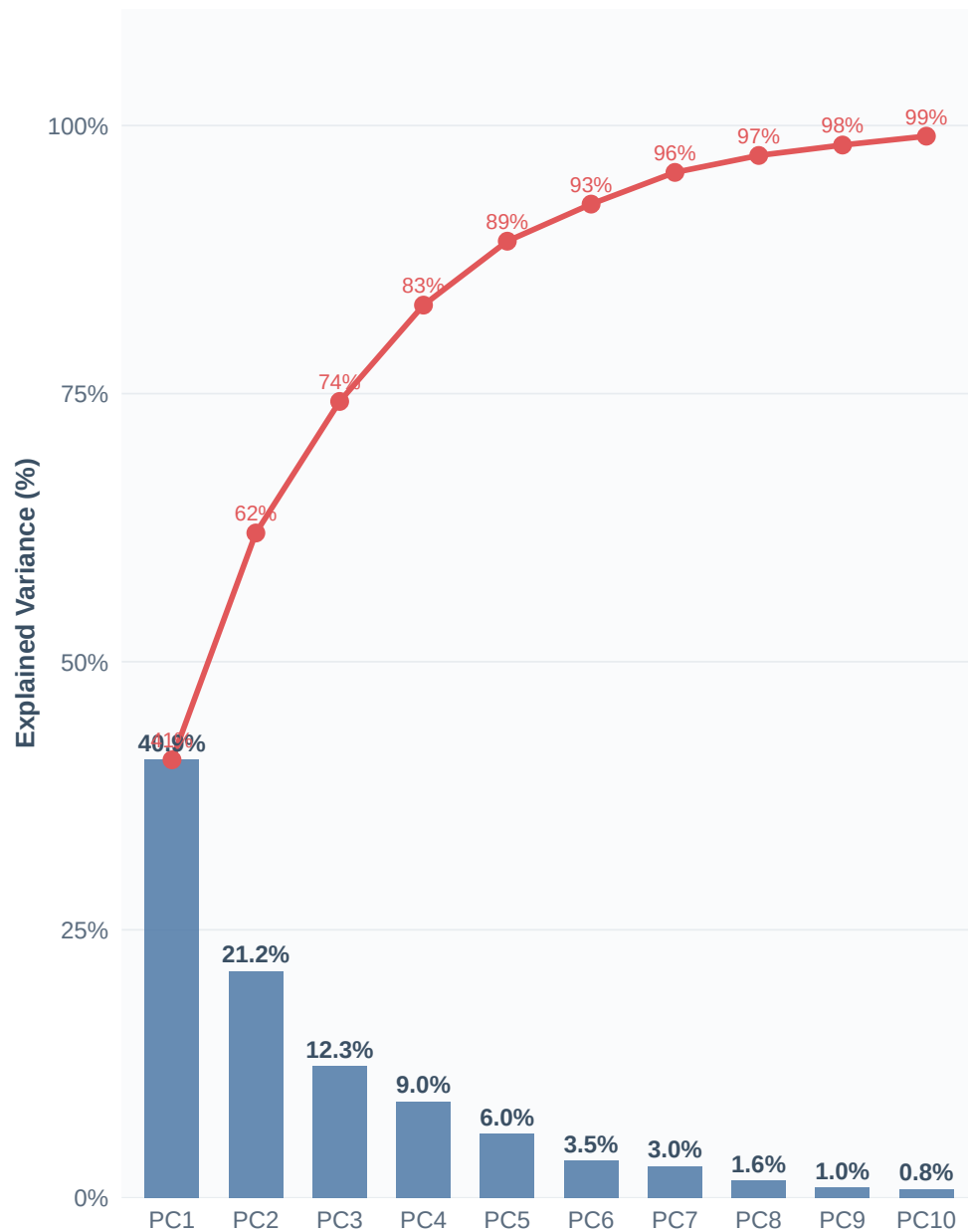


● Baseline (2) ● NS4 Fasting (18) ● RTDS Fasting (26)

Grey arrows: Baseline → NS4 Fasting → RTDS Fasting within each subject

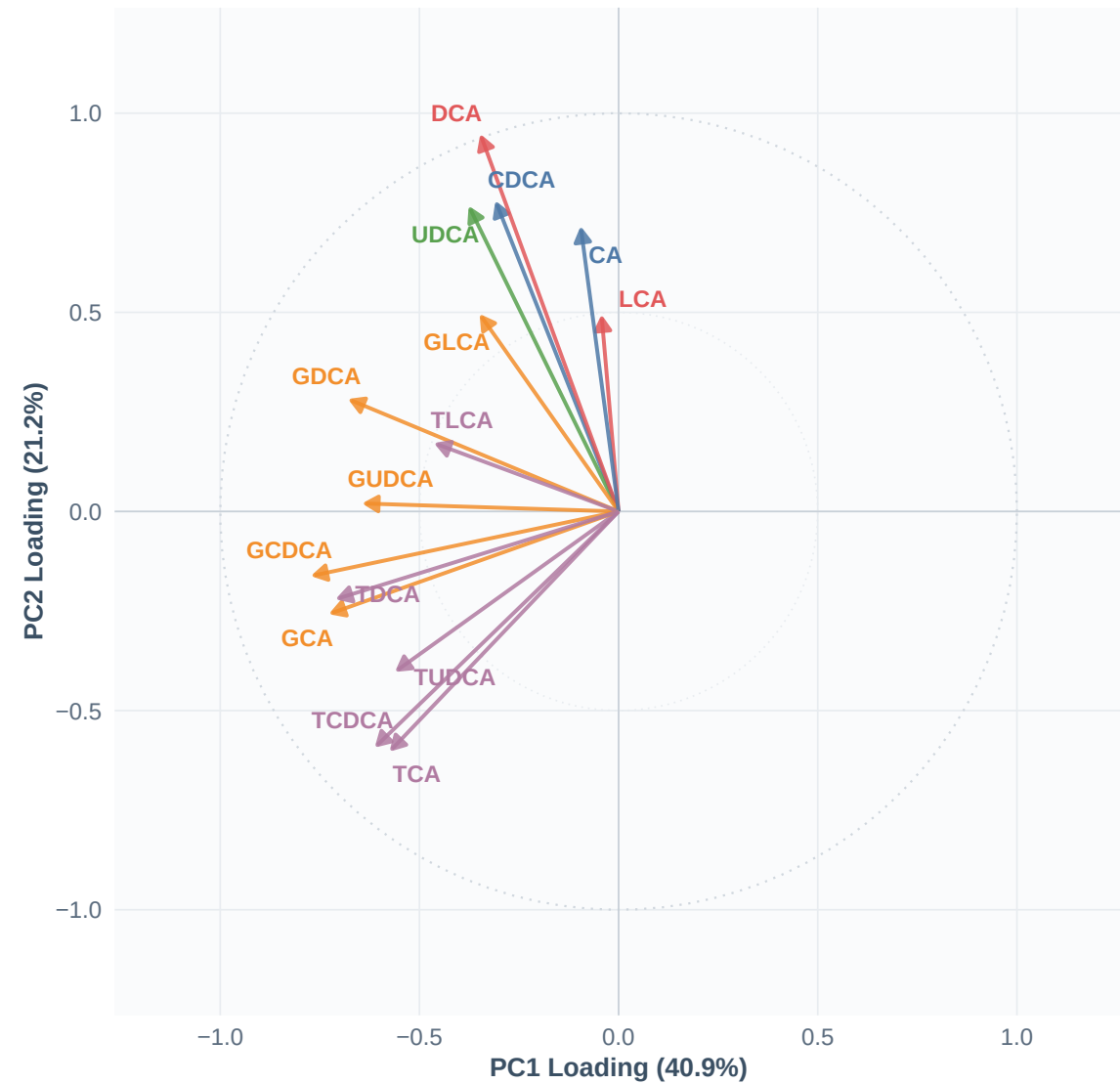
Variance Explained by Principal Components

Bars: individual variance | Red line: cumulative variance



PCA Loading Directions

Bile acids projected onto PC1 and PC2 (scaled to unit circle)

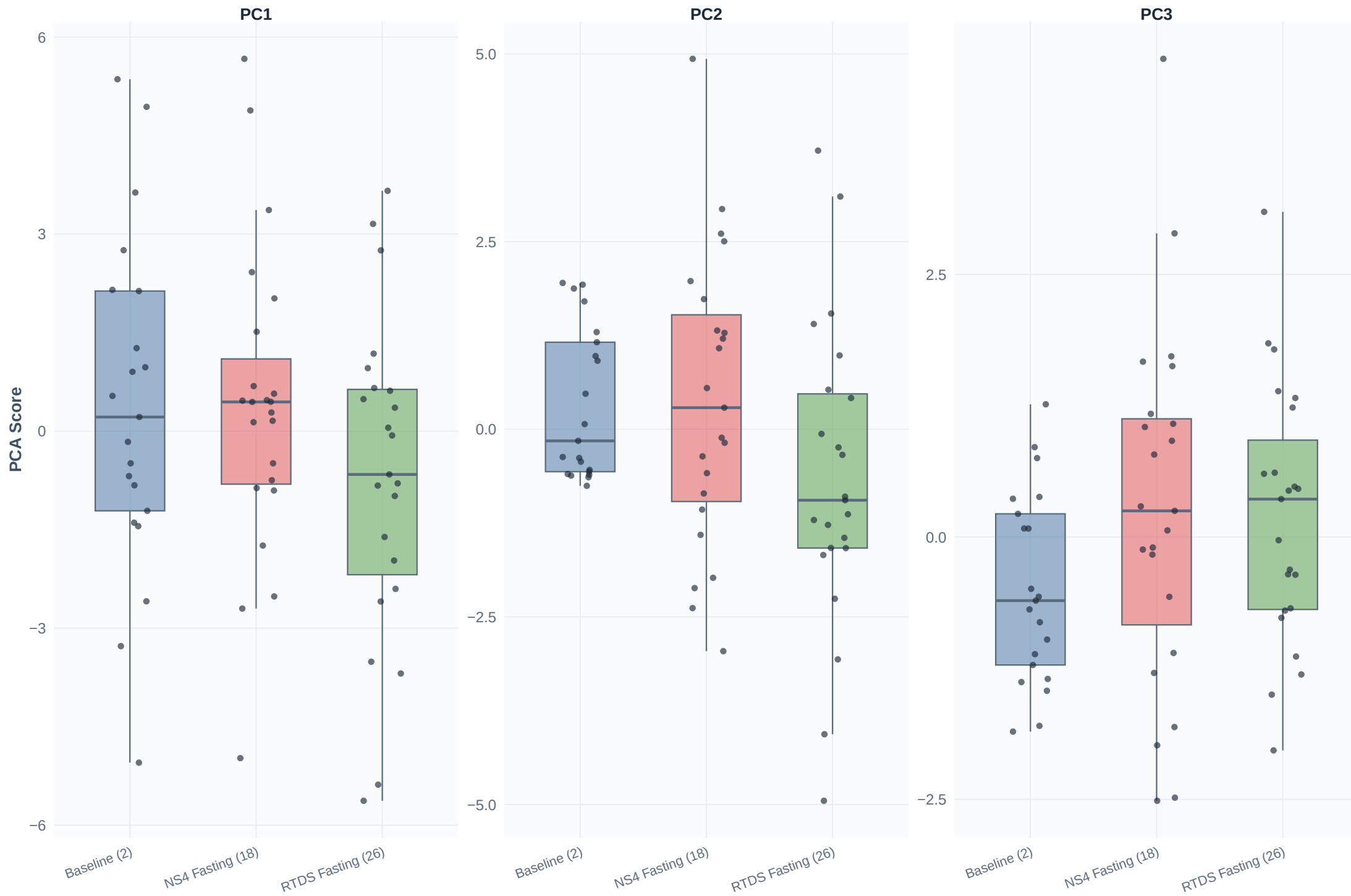


Bile Acid Class → Glycine-conj. → Primary → Secondary → Taurine-conj. → Tertiary

Arrow direction = loading direction | Colour = bile acid class

Score Distributions by Fasting Condition

First three principal components



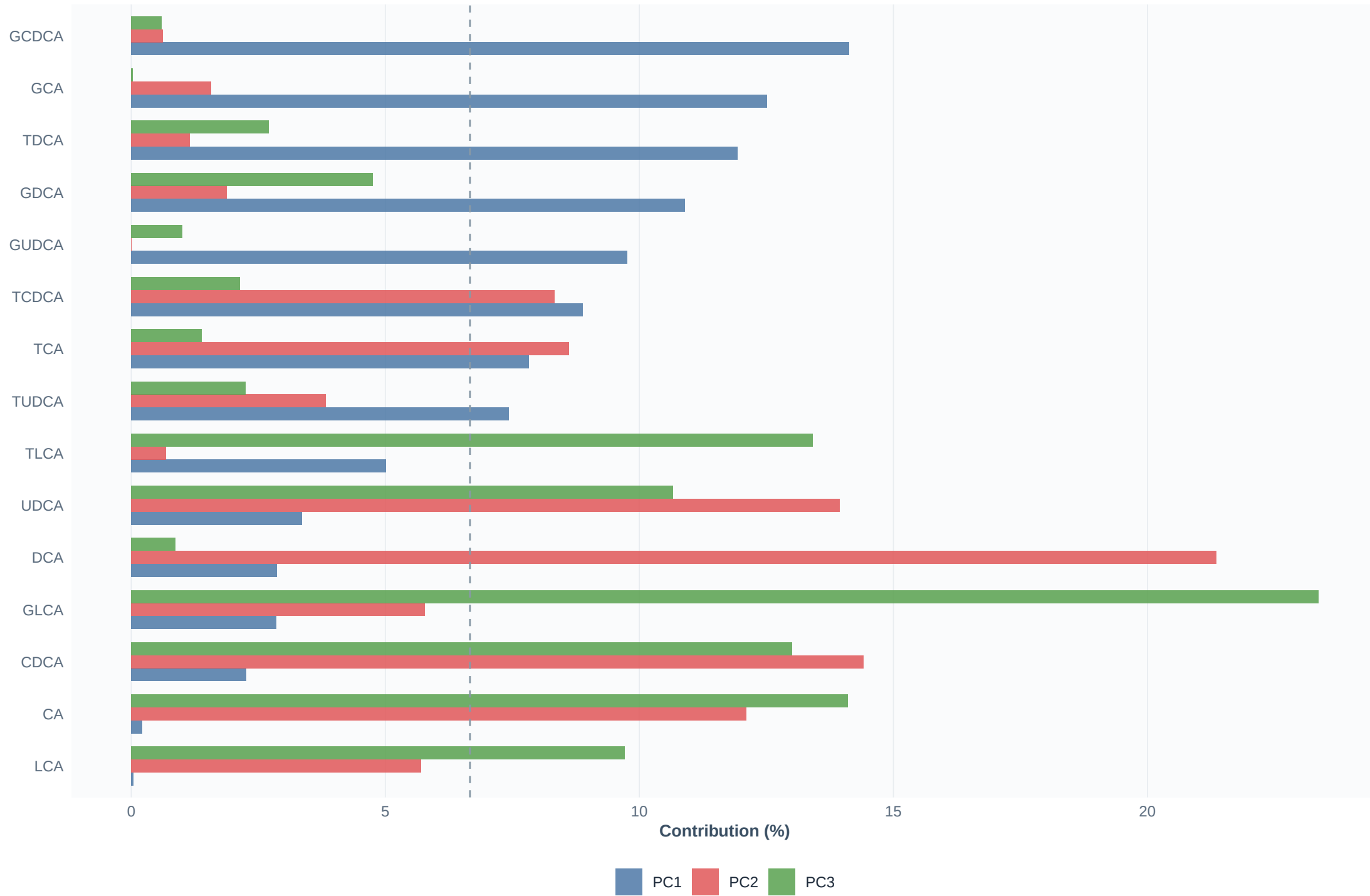
Top Bile Acid Loadings by Principal Component

Largest eight signed loadings (absolute value) for PC1–PC3



Variable Contributions to Principal Components

Dashed line = expected uniform contribution (6.7%)



PCA Output Files

PDF report: `fasting_only_analysis/outputs/Fasting_Only_PCA_Report.pdf`

CSV outputs (in

`fasting_only_analysis/outputs/pca`

/)

- `fasting_pca_scores.csv` — PC scores for each retained observation
- `fasting_pca_loadings.csv` — Variable loadings for all PCs
- `fasting_pca_variance.csv` — Variance explained per PC
- `fasting_pca_contributions.csv` — Variable contributions (squared loadings)
- `fasting_pca_input_log_complete_cases.csv` — Log-transformed input matrix
- `fasting_pca_missingness.csv` — Missing/non-positive row counts per compound
- `fasting_pca_verification_checks.csv` — Data audit trail

PNG plots (in

`fasting_only_analysis/outputs/pca/plots`

/)

- `pca_scores_plot.png` — Score plot with ellipses and centroids
- `pca_subject_trajectories.png` — Paired-subject movement in PCA space
- `pca_scree_plot.png` — Variance explained summary
- `pca_loadings_plot.png` — Loading directions with bile acid classes
- `pca_score_distributions.png` — Box plots of PC scores by condition
- `pca_top_loadings.png` — Top signed loadings for PC1–PC3
- `pca_variable_contributions.png` — Variable contribution percentages